Modelling Multi Species Occupancy Data

Effects of Habitat Type on the Occupancy and Detection of Mammal Communities

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Abstract

This project investigates how habitat type affects the presence and detectability of mammal species in a logged forest in Borneo. Camera trap data collected across 260 sites between 2014 and 2017 is used, covering 45 mammal species across five habitat types: trail, clearing, riparian, forest, and disturbed track. The challenge with camera trap data is that a species may go undetected at a site not because it is absent, but because the camera failed to photograph it. To address this, I fitted a multi-species occupancy model, which separates the probability that a species truly occupies a site from the probability that it is detected given it is there. The model is fitted using a Bayesian statistical framework, where inference is carried out using Markov Chain Monte Carlo sampling. Occupancy probability is modelled as a function of habitat type, while detection probability is modelled as a species-specific constant, because many species are absent from certain habitats entirely and the model therefore cannot estimate habitat-specific detection reliably. The results reveal which habitats support the highest mammal occupancy at the community level and identify which species show the strongest associations or avoidances with particular habitat types. I also discuss the limitations of the modelling approach and suggest directions for further research.

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1 Introduction and Background

1.1 Global Biodiversity Problem

Biodiversity is the variety of life on Earth across species and ecosystems, it is declining at a fast rate by human activity. The WWF Living Planet Report 2024, which tracked 34,836 populations across 5,495 vertebrate species, recorded an average decline of 73% in monitored wildlife populations between 1970 and 2020 (WWF, 2024).

Habitat loss and degradation are the main drivers of this problem. Approximately 75% of the Earth's land surface has been significantly altered by human activity, and natural ecosystems have declined by an average of 47% relative to their earliest estimated states (IPBES, 2019).In tropical regions, commercial logging and agricultural expansion are the dominant causes of forest loss, directly threatening the species that depend on these ecosystems.

1.2 Species Lost to Habitat Destruction

The consequences of unchecked habitat loss are visible in the extinction records of recent decades.

The Sumatran Rhinoceros in Borneo. Once common across Borneo, decades of commercial logging opened the forest interior to poaching and habitat conversion, reducing the population to just three captive individuals in Sabah by 2016. The last male, Tam, died in May 2019, and the last female, Iman, died in November 2019, rendering the subspecies locally extinct in Malaysian Borneo (National Geographic, 2019; Mongabay, 2019). The species was officially declared extinct in the wild in Malaysia in 2015.

The Spix's Macaw. The Spix's macaw, native to a small region of northeastern Brazil was declared extinct in the wild by the IUCN in 2019. The last known wild individual disappeared in 2000 after decades of deforestation for agriculture and livestock grazing destroyed the riparian gallery forests of the Caatinga biome on which the species depended. Only captive individuals survived, and reintroduction efforts began in 2022 with the release of 20 captive-bred birds. While this represents a rare conservation success, the species came within a single generation of permanent extinction.

By the time extinctions are officially recorded, the opportunity for intervention has already passed. This makes early detection of habitat-driven declines the main challenge of conservation biology. Camera trap surveys combined with occupancy modelling provide a systematic, statistically rigorous method of tracking which species are present in which habitats, long before populations reach the point of no return.

1.3 Camera Trap Survey

Sabah is a Malaysian state on Borneo known for its rainforests, diverse wildlife like orangutans or elephants. The Stability of Altered Forest Ecosystems (SAFE) project is one of the world's largest ecological experiments, based in Sabah, Malaysian Borneo. This project explores the modelling of

the camera-trap data collected from the experiment.

A camera trap is a remotely activated digital camera used for capturing photos or videos of wildlife, triggered by motion or infrared sensors. These devices are used for research, conservation, and monitoring, and they can be left in nature for weeks or months, providing data on animal populations.

Camera trap surveys are a standard tool for monitoring wildlife in dense environments because they generate large datasets of species detections across many sites and time periods. However, cameras can fail to photograph a mammal during a study period, so species may go undetected at a site even when they are not absent. Therefore, we cannot measure the true presence of the species from raw data alone. Occupancy modelling provides a statistical framework that separates the ecological process from the observation process. The core model, introduced by MacKenzie (2002), uses repeated surveys at each site to estimate both occupancy probability (psi) and detection probability (p) simultaneously. This can be implemented in R using the spOccupancy package (Doser 2022), which allows the entire mammal community to be modelled jointly, with species borrowing statistical strength from one another through a shared hierarchical prior structure.

1.4 Motivation

This project is motivated by ecological and statistical considerations that are directly relevant to conservation science. Fighting against biodiversity loss by providing evidence based solutions is the best way to protect species. This project estimates the probability that each species uses different habitat types. By doing this, it helps show which habitats act as refuges for sensitive species and which ones function more like ecological sinks. At the same time, it estimates how likely animals are to be detected given they are present at a site. This makes it easier to fairly compare results across habitats that vary in vegetation density, terrain, and how animals move through them.

2 Mathematical Framework

2.1 The Latent Occupnacy State

A fundamental challenge in camera trap ecology is that the true presence or absence of a species at a site is never directly observed. Even if a species is present at a site, we don't know if the camera will photograph it during the survey period. To handle this, occupancy models introduce an unobserved binary variable that represents the true biological state of each site.

For species k at site i, we define:

$$z_{k,i} = \begin{cases} 1 & \text{if species } k \text{ truly occupies site } i \\ 0 & \text{if species } k \text{ is truly absent from site } i \end{cases} \tag{1}$$

The variable $z_{k,i}$ is latent because we have never observed it directly. What we observe instead is the detection history $y_{k,i,t}$ across repeated survey occasions $t = 1, \ldots, T$, which is a reflection of the true state. The latent state $z_{k,i}$ is modelled as a Bernoulli random variable:

$$z_{k,i} \sim \text{Bernoulli}(\psi_{k,i}) \tag{2}$$

where $\psi_{k,i} \in [0, 1]$ is the *occupancy probability* which is the probability that species k truly occupies site i. The Bernoulli distribution is appropriate here because $z_{k,i}$ takes only two values (present or absent) and the outcome at each site is assumed to be independent of outcomes at other sites.

2.2 The Ecological Process — Occupancy

The occupancy probability $\psi_{k,i}$ is modelled as a function of habitat type. Since $\psi_{k,i}$ is a probability bounded between 0 and 1, it cannot be modelled directly with a linear predictor, which is unbounded. Instead, we apply the *logit link function*:

$$\text{logit}(\psi_{k,i}) = \log\left(\frac{\psi_{k,i}}{1 - \psi_{k,i}}\right) \tag{3}$$

This transforms $\psi_{k,i}$ from the probability scale $[0, 1]$ to the real line $(-\infty, \infty)$, on which a linear predictor can be freely defined. The inverse of the logit function converts back to the probability scale:

$$\psi_{k,i} = \frac{1}{1 + e^{-\eta_{k,i}}} \tag{4}$$

where $\eta_{k,i}$ is the linear predictor. The full occupancy model with habitat type as the covariate is:

$$\text{logit}(\psi_{k,i}) = \beta_{k,0} + \beta_{k,1}\, x_{\text{Clearing},i} + \beta_{k,2}\, x_{\text{Riparian},i} + \beta_{k,3}\, x_{\text{Forest},i} + \beta_{k,4}\, x_{\text{Disturbed Track},i} \tag{5}$$

where $x_{h,i} \in \{0, 1\}$ is a dummy variable equal to 1 if site i belongs to habitat h and 0 otherwise. Trail is the *reference level*, absorbed into the intercept $\beta_{k,0}$. The coefficients $\beta_{k,1}, \ldots, \beta_{k,4}$ represent the change in log-odds of occupancy when moving from a Trail site to each respective habitat type, for species k. The occupancy probability at each habitat is therefore:

$$\text{Trail:} \quad \text{logit}(\psi_{k,i}) = \beta_{k,0} \tag{6}$$

$$\text{Clearing:} \quad \text{logit}(\psi_{k,i}) = \beta_{k,0} + \beta_{k,1} \tag{7}$$

$$\text{Riparian:} \quad \text{logit}(\psi_{k,i}) = \beta_{k,0} + \beta_{k,2} \tag{8}$$

$$\text{Forest:} \quad \text{logit}(\psi_{k,i}) = \beta_{k,0} + \beta_{k,3} \tag{9}$$

$$\text{Disturbed Track:} \quad \text{logit}(\psi_{k,i}) = \beta_{k,0} + \beta_{k,4} \tag{10}$$

A positive value of $\beta_{k,j}$ indicates that species k is more likely to occupy habitat j than a Trail site. A negative value indicates an avoidance of that habitat relative to Trail.

2.3 The Observation Process — Detection

Given that species k is present at site i, we model the detection of that species during survey occasion t as:

$$y_{k,i,t} \mid z_{k,i} \sim \text{Bernoulli}(z_{k,i} \cdot p_k) \tag{11}$$

where $y_{k,i,t} \in \{0, 1\}$ is the observed detection (1 if the species was photographed, 0 otherwise) and p_k is the *detection probability* — the probability the camera records species k during a single survey occasion, given the species is present. The term $z_{k,i}$ acts as a switch: if the species is truly absent ($z_{k,i} = 0$), detection is impossible regardless of p_k. If the species is present ($z_{k,i} = 1$), detection occurs with probability p_k. Detection probability is modelled as a **species-specific constant** meaning it does not vary by habitat type or survey occasion:

$$\text{logit}(p_k) = \alpha_{k,0} \tag{12}$$

The α coefficients are entirely separate from the β coefficients. A species may be highly detectable when present (p_k large) but occupy very few sites ($\psi_{k,i}$ small). Without separating the two sub-models, these two sources of variation in detection counts would be confounded in the raw data.

2.4 Dummy Variable Coding and the Design Matrix

The five habitat categories cannot be entered directly into a regression model as text labels. Instead, they are encoded using *dummy variables*. Given $H = 5$ habitat types with Trail as the reference level, we will need $H - 1 = 4$ dummy variables. The design matrix applies only to the occupancy sub model. The design matrix $\mathbf{X}$ has one row per habitat type and one column per coefficient:

$$\mathbf{X} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix} \tag{13}$$

The linear predictor for species k at habitat h is computed as the dot product of the species' coefficient vector $\boldsymbol{\beta}_k = (\beta_{k,0}, \beta_{k,1}, \beta_{k,2}, \beta_{k,3}, \beta_{k,4})^\top$ and the corresponding row of $\mathbf{X}$:

Table 1: Design matrix $\mathbf{X}$ for the five habitat types. Trail is the reference level. Each row represents one habitat type as a vector of dummy variables.

	Intercept	Clearing	Riparian	Forest	Disturbed Track
Trail	1	0	0	0	0
Clearing	1	1	0	0	0
Riparian	1	0	1	0	0
Forest	1	0	0	1	0
Disturbed Track	1	0	0	0	1

$$\text{logit}(\psi_{k,i}) = \boldsymbol{\beta}_k^\top \mathbf{x}_i = \sum_{j=0}^{4} \beta_{k,j}\, x_{j,i} \tag{14}$$

For example, for a Riparian site where $\mathbf{x}_i = (1, 0, 1, 0, 0)^\top$:

$$\boldsymbol{\beta}_k^\top \mathbf{x}_{\text{Riparian}} = \beta_{k,0} \cdot 1 + \beta_{k,1} \cdot 0 + \beta_{k,2} \cdot 1 + \beta_{k,3} \cdot 0 + \beta_{k,4} \cdot 0 = \beta_{k,0} + \beta_{k,2} \tag{15}$$

This confirms that the linear predictor for Riparian consists of only the intercept and the Riparian coefficient. In matrix form across all sites simultaneously:

$$\text{logit}(\boldsymbol{\psi}_k) = \mathbf{X}\boldsymbol{\beta}_k \tag{16}$$

2.5 The Marginal Likelihood

The latent occupancy state $z_{k,i}$ is never observed directly. To construct the likelihood of the observed data $\mathbf{Y}$, we must *marginalise* the unknown $z_{k,i}$. For species k at site i, the marginal likelihood is obtained by summing over the two possible values of $z_{k,i}$:

$$\mathcal{L}_{k,i} = \psi_{k,i} \prod_{t=1}^{T} p_k^{y_{k,i,t}} (1 - p_k)^{1 - y_{k,i,t}} + (1 - \psi_{k,i}) \cdot \mathbb{1}\left[\sum_{t=1}^{T} y_{k,i,t} = 0\right] \tag{17}$$

Term 1 — Species Present

The first term represents the scenario where species k truly occupies site i. The probability of occupancy is $\psi_{k,i}$. Given the species is present, the detection history across all T occasions is modelled as a product of independent Bernoulli trials. The exponent $y_{k,i,t}$ acts as a selector: when $y_{k,i,t} = 1$ (species detected), the term contributes $p_k^1(1 - p_k)^0 = p_k$; when $y_{k,i,t} = 0$ (species not detected), the term contributes $p_k^0(1-p_k)^1 = 1-p_k$. The full product therefore gives the probability of the exact observed detection pattern across all occasions.

Term 2 — Species Absent

The second term represents the scenario where species k is truly absent from site i. The probability of absence is $(1 - \psi_{k,i})$. If the species is absent, it cannot be detected, so this term can only

contribute when the species was *never* detected across all occasions. The indicator function $\mathbb{1}[\cdot]$ defines this:

$$\mathbb{1}\left[\sum_{t=1}^{T} y_{k,i,t} = 0\right] = \begin{cases} 1 & \text{if the species was never detected (all } y_{k,i,t} = 0) \\ 0 & \text{if the species was detected at least once} \end{cases} \tag{18}$$

The moment a species is detected even once at a site, Term 2 vanishes entirely. The species must have been present, so only Term 1 contributes to the likelihood.

2.6 Prior Distributions

The community-level means and variances are themselves unknown parameters and require prior distributions. The defaults of the `spOccupancy` package use weakly informative priors:

$$\mu_{\beta_j} \sim \mathcal{N}(0,\ 2.72) \tag{19}$$

$$\mu_{\alpha_0} \sim \mathcal{N}(0,\ 2.72) \tag{20}$$

$$\sigma^2_{\beta_j} \sim \text{Inverse-Gamma}(0.1,\ 0.1) \tag{21}$$

$$\sigma^2_{\alpha_0} \sim \text{Inverse-Gamma}(0.1,\ 0.1) \tag{22}$$

The prior $\mathcal{N}(0, 2.72)$ on the logit scale is *weakly informative*. To understand why, consider converting a value of ± 2 on the logit scale back to a probability:

$$\psi = \frac{1}{1+e^{-2}} \approx 0.88, \qquad \psi = \frac{1}{1+e^{+2}} \approx 0.12 \tag{23}$$

A standard deviation of 2.72 on the logit scale therefore spans roughly the range $[0.05, 0.95]$ on the probability scale, covering nearly all possible values. This shows prior ignorance about which habitats are better or worse before seeing the data. All inferences are therefore driven by the observed detection histories rather than prior assumptions. The Inverse-Gamma$(0.1, 0.1)$ prior on the community variances is also weakly informative, placing most prior mass on a wide range of values and allowing the data to determine how much species vary from the community mean.

2.7 The Full Posterior Distribution

Combining the likelihood, the priors via Bayes' theorem gives the full joint posterior distribution:

$$P(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{z}, \boldsymbol{\mu}, \boldsymbol{\sigma}^2 \mid \mathbf{Y}) \propto \underbrace{\prod_{k=1}^{K}\prod_{i=1}^{I} \mathcal{L}_{k,i}}_{\text{likelihood}} \times \underbrace{P(\boldsymbol{\beta} \mid \boldsymbol{\mu}_\beta, \boldsymbol{\sigma}^2_\beta)}_{\text{prior on } \boldsymbol{\beta}} \times \underbrace{P(\boldsymbol{\alpha} \mid \mu_{\alpha_0}, \sigma^2_{\alpha_0})}_{\text{prior on } \boldsymbol{\alpha}} \times \underbrace{P(\boldsymbol{\mu}, \boldsymbol{\sigma}^2)}_{\text{hyperpriors}} \tag{24}$$

The double product $\prod_{k=1}^{K}\prod_{i=1}^{I}$ applies only to the likelihood, which depends on both the species index k and the site index i. The priors and hyperpriors sit outside the product because they are defined at the parameter level. The posterior is the central object of inference. It represents the distribution of all parameters given the observed camera trap data, combining the information in the likelihood with the regularisation provided by the hierarchical priors. This posterior has no closed-form solution. The parameter space has hundreds of dimensions, five occupancy coefficients

per species across $K = 45$ species, one detection intercept per species, community means and variances, the latent occupancy states $z_{k,i}$ for every species at every site. We therefore require the Markov Chain Monte Carlo methods.

3 Computational Methods

In this section we describe the methods used to obtain posterior estimates of occupancy probability for each species across each habitat type, and detection probability for each species. The full code for this project is available on GitHub. The code also includes diagnostics, which show that the model works perfectly.

`https://github.com/Zuhan-Shahria/Modelling-Multi-Species-Occupancy-Data.git`

3.1 Why We Need MCMC

The posterior distribution derived in Section 2.7 cannot be solved analytically. To understand why, consider that the parameter space includes five occupancy coefficients and one detection intercept for each of the 45 species, community means and variances for each coefficient, and a latent occupancy state $z_{k,i}$ for every species at every site. Computing the posterior exactly would require integrating over all of these simultaneously, which is difficult. Markov Chain Monte Carlo (MCMC) is a computational method that solves this problem by *sampling* from the posterior rather than solving for it exactly. Instead of writing down a formula for the posterior, MCMC explores it by taking a long sequence of steps through the parameter space, spending more time in regions of high probability. After many steps, the collection of visited positions approximates the posterior distribution. The posterior mean and credible interval can be computed from these samples.

3.2 The Gibbs Sampler

The specific MCMC algorithm used by the `spOccupancy` package is a *Gibbs sampler*. Rather than proposing a random move and deciding whether to accept it, the Gibbs sampler updates one parameter at a time by sampling directly from its *full conditional distribution* which is the distribution of that parameter given the current values of all other parameters. This is more efficient than the random approach because each update is guaranteed to produce a valid sample. One complete Gibbs iteration in this model is the following:

1. Update all occupancy coefficients $\beta_{k,j}$ for every species and habitat, given the current values of the latent states $z_{k,i}$, the community means μ_{β_j}, and the community variances $\sigma^2_{\beta_j}$.
2. Update all detection intercepts $\alpha_{k,0}$ for every species, given the current values of $z_{k,i}$, μ_{α_0}, and $\sigma^2_{\alpha_0}$.
3. Update the latent occupancy state $z_{k,i}$ for every species at every site. For sites where the species was detected at least once, $z_{k,i}$ must equal 1. For sites with zero detections, the sampler weighs the probability of true absence against presence with missed detections, using the current values of $\psi_{k,i}$ and p_k.
4. Update the community means μ_{β_j} and μ_{α_0}, given all current species-level coefficients.
5. Update the community variances $\sigma^2_{\beta_j}$ and $\sigma^2_{\alpha_0}$, given all current species-level coefficients and the updated community means.

Each complete cycle through these steps is one MCMC iteration. The sampler repeats this process for 10,000 iterations per chain.

3.3 Burn-in, Thinning and Multiple Chains

There are three main settings that control the quality of the posterior samples.

Burn-in refers to the initial samples that are discarded. When the sampler starts, it begins from an arbitrary position in the parameter space and takes some number of steps before it finds the high-probability region. Samples taken during this period do not represent the posterior reliably. In this project, the first 5,000 iterations of each chain are discarded as burn-in, leaving the remaining 5,000 iterations for inference.

Thinning means keeping only every nth sample to reduce autocorrelation. Consecutive MCMC steps tend to be similar to each other, so keeping every step gives redundant information. Keeping every 5th sample gives a more independent set of draws. Applied to 5,000 post-burn-in samples, thinning by 5 produces 1,000 draws per chain.

Multiple chains means running the sampler several times from different random starting positions. This project uses 3 chains, each producing 1,000 posterior draws, giving a total of 3,000 effective posterior draws. The purpose of running multiple chains is to check convergence.

After fitting the model, it is important to check that the MCMC has actually converged to the true posterior rather than getting stuck in one region of the parameter space. The **Gelman-Rubin $\hat{R}$ statistic** (Gelman and Rubin, 1992) compares the variation *within* each chain to the variation *between* chains. If all three chains have converged to the same distribution, within-chain and between-chain variation should be similar and $\hat{R}$ will be close to 1.0. Values below 1.1 are considered acceptable evidence of convergence. A value much larger than 1.1 indicates the chains have not mixed and more samples are needed.

3.4 Posterior Summarisation

Once the 3,000 posterior draws have been obtained, they are used to compute summaries for each parameter of interest.
The **posterior mean** is the average value across all 3,000 draws. For occupancy probability at habitat h for species k, this is:

$$\hat{\psi}_{k,h} = \frac{1}{3000} \sum_{d=1}^{3000} \psi_{k,h}^{(d)} \tag{25}$$

where $\psi_{k,h}^{(d)}$ is the occupancy probability computed from the dth posterior draw by applying the inverse logit to the linear predictor $\boldsymbol{\beta}_k^{(d)\top} \mathbf{x}_h$. The **95% credible interval** is obtained by sorting all 3,000 draws from smallest to largest and taking the values at the 2.5th and 97.5th percentiles:

$$\text{CRI}_{95\%} = \left[Q_{0.025}\left(\psi_{k,h}^{(1:3000)}\right),\ Q_{0.975}\left(\psi_{k,h}^{(1:3000)}\right) \right] \tag{26}$$

A narrow interval means the estimate is precise. A wide interval means there is substantial uncertainty, most likely because few sites of that habitat type were surveyed for that species. The same procedure is applied to detection probability using the draws of $\alpha_{k,0}$.

4 Data and Preparation

This section describes the raw data used in the project, the steps taken to clean and prepare it, and how it was structured into the format required by the occupancy model.

4.1 Study Area and Data Collection

The data came from the Stability of Altered Forest Ecosystems (SAFE) project, a large-scale ecological experiment based in Sabah, Malaysian Borneo. The landscape is a heavily logged lowland rainforest mosaic containing continuous logged forest, forest fragments of different sizes, oil palms, riparian corridors. Camera traps were deployed across 260 sites between 2014 and 2017, with each camera active for a deployment period typically lasting six to eight weeks. The data were provided as two separate CSV files: a trap effort dataset recording the details of each deployment, and a mammal detection dataset recording every photograph triggered by an animal.

4.2 The Datasets

The **trap effort dataset** contains one row per camera deployment. The key columns used in this project are the site identifier (`Site_ID`), the deployment start and end dates (`Date_On` and `Date_Off`), and the habitat feature recorded at each site (`Habitat_Feature`).

The **mammal detection dataset** contains one row per photograph. The key columns are the site identifier (`Location`, renamed to `Site_ID`), the date of the photograph (`Date`), the species identified (`Species`), and a contact classification column (`New_Contact`) indicating whether the photograph represents an independent detection event or a repeat photograph within the same encounter.

Both datasets required date columns to be converted from character strings in day/month/year format to R `Date` objects using the `dmy()` function from the `lubridate` package before any temporal calculations could be performed.

4.3 Filtering Independent Detections

Camera traps fire multiple times per animal encounter due to burst photography, meaning a single visit by one animal can generate consecutive photographs. Including all photographs would inflate detection counts and violate the assumption that detections within a survey occasion are independent. Therefore, only rows where `New_Contact == "New Contact"` were kept, keeping one record per independent detection event and discarding all repeat photographs within the same encounter. This step reduced the detection dataset to a single observation per independent animal contact.

4.4 Matching Detections to Deployments

To ensure each detection was attributed to the correct trap deployment, the filtered detection dataset was joined to the trap effort dataset using `Site_ID` as the key, and then filtered to retain only detections where the photograph date fell within the active deployment window:

$$\texttt{Date_On} \leq \texttt{Date} \leq \texttt{Date_Off}$$

This step is necessary because some sites were deployed multiple times across different years. Without this filter, a detection from 2016 could be incorrectly matched to a 2014 deployment at the same site.

4.5 Defining Survey Occasions

The occupancy model requires detection data to be organised into discrete repeated survey occasions. Each deployment was divided into weekly occasions by computing how many days had elapsed since the camera was switched on and dividing by seven:

$$\text{week} = \left\lfloor \frac{\text{Date} - \text{Date_On}}{7} \right\rfloor + 1$$

For example, a detection on day 0 to day 6 of a deployment falls in week 1, a detection on day 7 to day 13 falls in week 2, and so on. This produces the repeated survey structure the model requires to separate occupancy from detection. The maximum number of weeks across all deployments determines the third dimension of the detection array.

4.6 Habitat Classification

The raw trap effort dataset contained 18 distinct habitat labels, many of which were ecologically similar or present at only one or two sites. A covariate with 18 levels would produce unreliable coefficient estimates for the rarest categories. The habitat labels were therefore consolidated into five ecologically meaningful groups using a two-step process.

In the first step, sites recorded as `Random` were reclassified using their `Block` label. The Random label describes the camera placement strategy, the field team chose the location without targeting any specific habitat feature rather than the physical environment the camera was sitting in. The Block column records the forest management unit the site belongs to and therefore provides a meaningful ecological description. Replacing Random with Block ensures these sites contribute useful information to the model rather than forming an uninterpretable category.

In the second step, all remaining habitat labels were merged into five categories using ecological reasoning:

- **Trail** — Trail, Trail (ridge), Trail convergence, and Random/trail. All represent linear pathways used for animal and human movement.
- **Clearing** — Clearing and Forest clearing. Both represent open habitats created by disturbance.
- **Riparian** — Riparian, Riparian logged forest, Riparian trail, and Logged Forest Control River. All are water-adjacent habitats.
- **Forest** — Forest fragment, Forest Block, Oil palm, Logged Forest Control, and Oil palm Riparian Reserve. All represent remnant or interior forest conditions.
- **Disturbed Track** — Logging road and Skid trail. Both are linear disturbance features created by logging machinery.

Trail was set as the reference level, so all habitat coefficients are interpreted as deviations from Trail. Sites with blank or missing habitat labels after this process were excluded from the analysis, as they could not be assigned to any meaningful ecological category.

4.7 Building the Detection Array

The detection data were structured into a three-dimensional array $\mathbf{Y}$ with dimensions species $\times$ sites $\times$ weeks. Each cell $y_{k,i,t}$ was set to 1 if species k was detected at site i during week t, and 0 otherwise. The array was initialised as all zeros and then populated using a nested loop that located the correct species, site, and week indices for each independent detection record. The final dimensions of the array were $45 \times I \times T$, where I is the number of clean sites after habitat filtering and T is the maximum number of weekly occasions observed across all deployments.

4.8 Handling Missing Data

Two types of missing data required different treatments.

Missing habitat labels were handled by exclusion. The `good_sites` filter removed any site where `Habitat_Feature` was `NA` or a blank string after the merging step. These sites were dropped from the detection array, the site covariate list, and the list of clean site identifiers before model fitting.

Unsampled occasions were handled by NA masking. Not every site was actively surveyed during every week of the study period. A week where the trap was not yet deployed or had already been collected should not be treated as a week with zero detections, because a zero detection requires an active camera. An effort matrix was constructed recording which site-week combinations had an active trap, and cells corresponding to unsampled occasions were set to `NA` in both the detection array and the detection covariate matrix. The `spOccupancy` package treats `NA` cells as missing rather than as non-detections, ensuring they do not contribute incorrectly to the likelihood.

5 Results

This section presents the posterior estimates of occupancy probability per species per habitat, detection probability per species, community-level occupancy across habitat types, and the heatmap summarising habitat associations across the full mammal community.

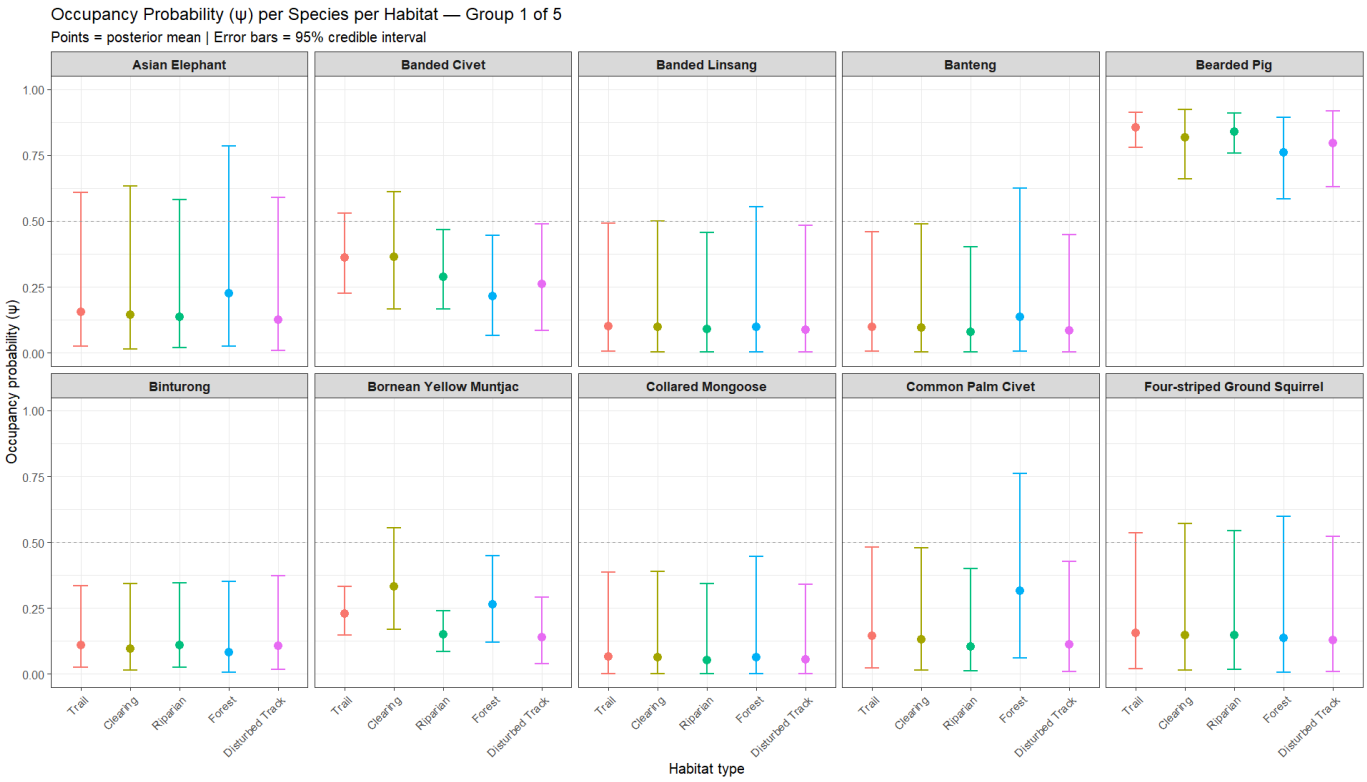
5.1 Occupancy Probability per Species per Habitat

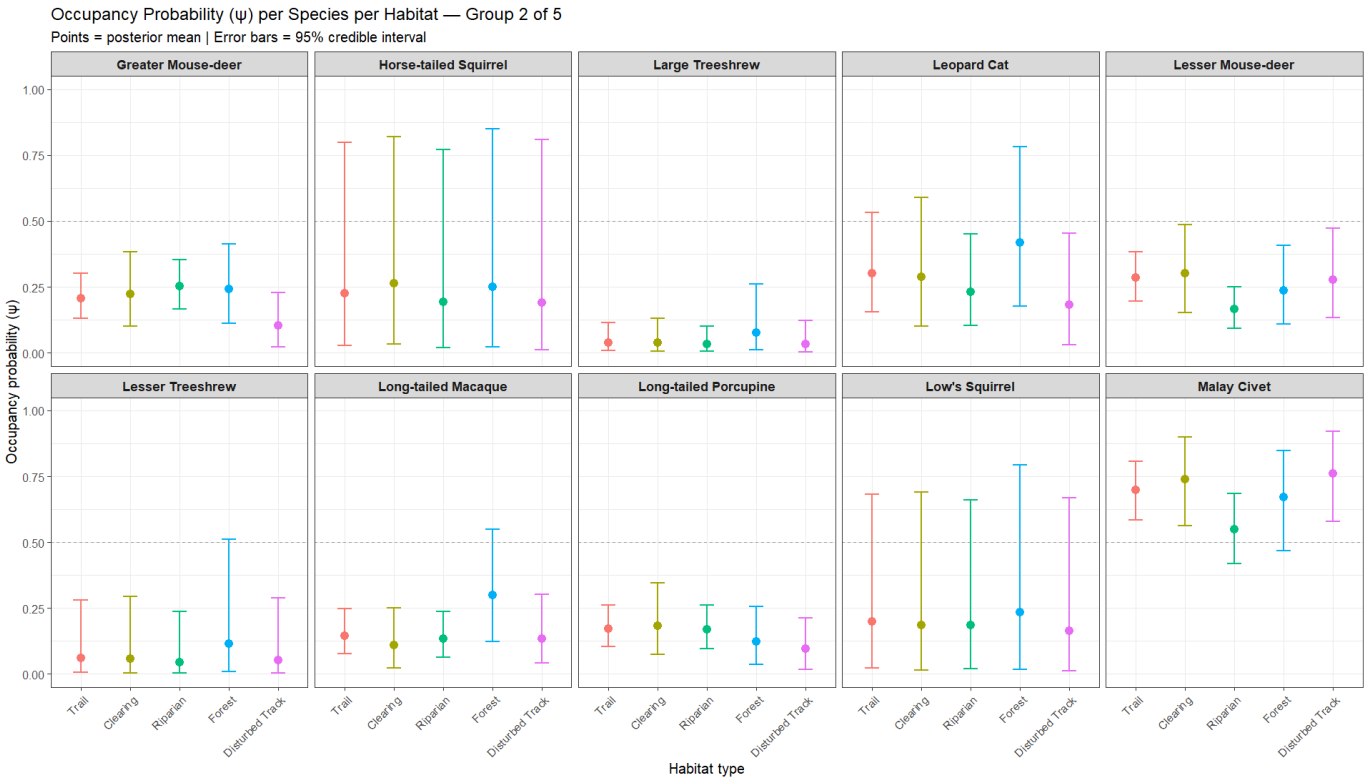
Figure 1 shows the posterior mean occupancy probability $\hat{\psi}_{k,h}$ and 95% credible interval for each of the 45 species across the five habitat types. The y-axis is standardised to $[0, 1]$ across all panels to allow direct comparison of effect sizes between species.

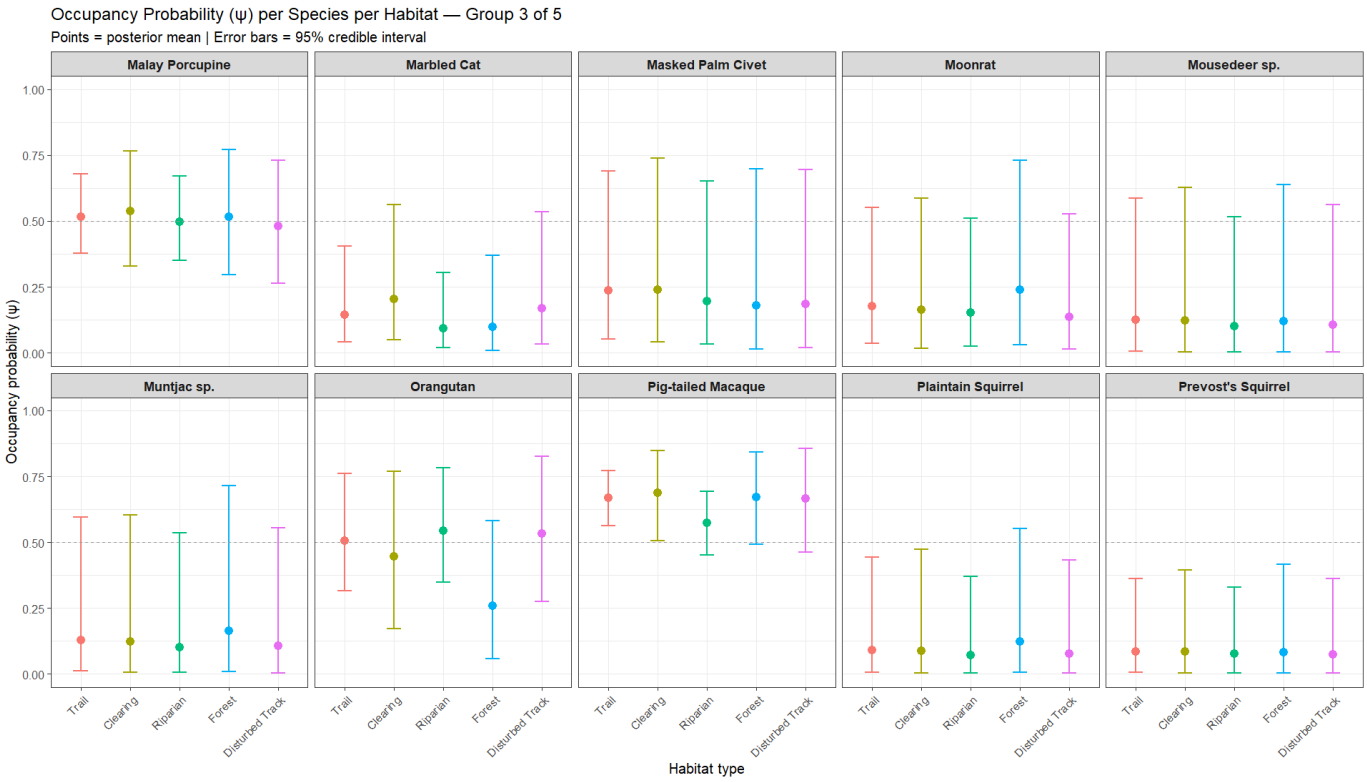
Several broad patterns are visible. The majority of species have posterior mean occupancy probabilities below 0.5 across all habitats, indicating that most species occupy a relatively small proportion of sites in this landscape. However, a small number of species stand out as exceptions. Bearded Pig shows notably high occupancy across all five habitat types with posterior means close to or exceeding 0.75, suggesting it is the most widespread mammal in the dataset with little preference for any particular habitat. Malay Civet, Pig-tailed Macaque, Orangutan, Red Muntjac, and Malay Porcupine similarly show relatively high and consistent occupancy across habitats compared to the rest of the community.

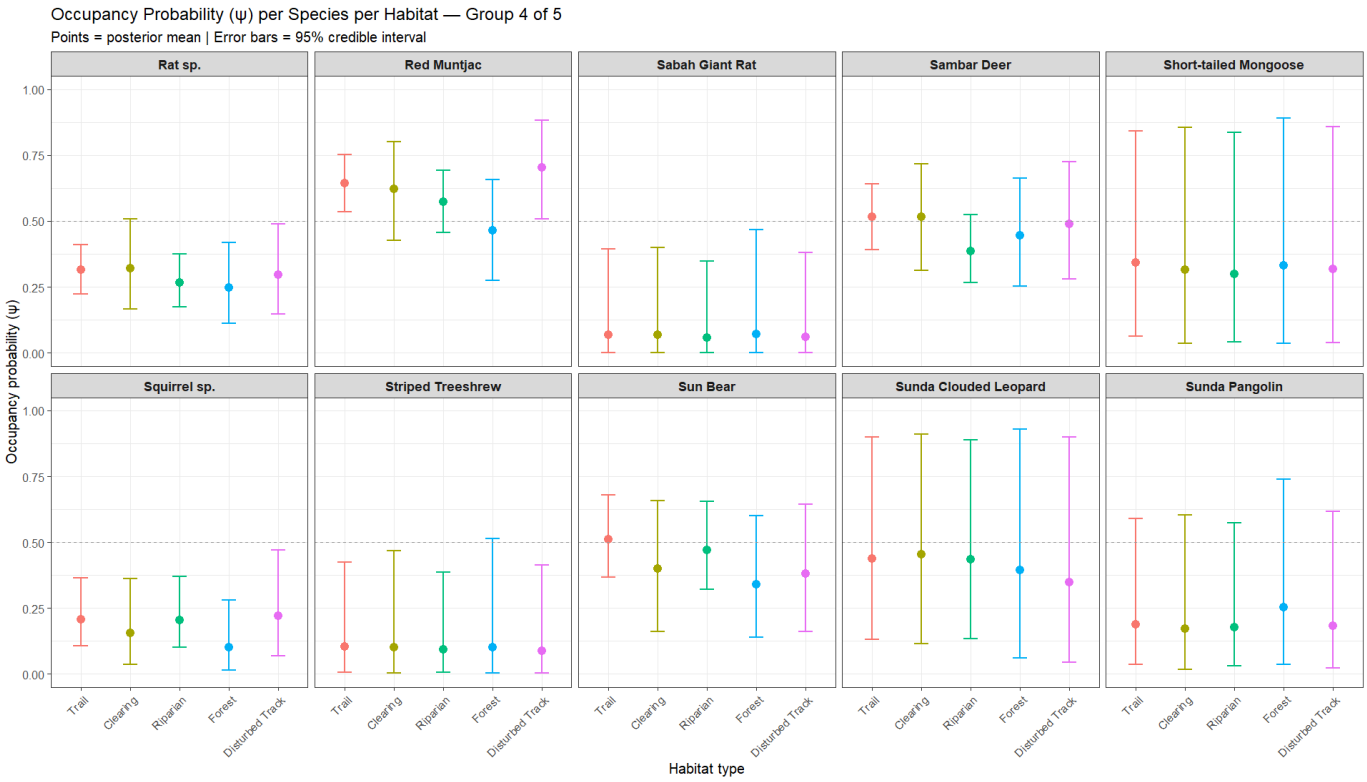
Many species show wide credible intervals, particularly in the Disturbed Track and Clearing categories. This reflects the relatively small number of camera deployments in these habitat types and is consistent with what would be expected because fewer sites means less data and therefore greater posterior uncertainty. By contrast, Trail and Riparian categories generally show narrower intervals, reflecting the larger number of deployments in these habitats.

For most species the posterior means are broadly similar across habitats, with overlapping credible intervals. This suggests that for the majority of the community, the data do not provide strong evidence of habitat-specific occupancy differences on the logit scale, though this is partly a consequence of the relatively small number of sites in some habitat categories.


Occupancy Probability (ψ) per Species per Habitat — Group 1 of 5
Points = posterior mean | Error bars = 95% credible interval
Asian Elephant
Banded Civet
Banded Linsang
Banteng
Bearded Pig
Binturong
Bornean Yellow Muntjac
Collared Mongoose
Common Palm Civet
Four-striped Ground Squirrel
Occupancy probability (ψ)
1.00
0.75
0.50
0.25
0.00
Trail
Clearing
Riparian
Forest
Disturbed Track
Habitat type


Occupancy Probability (ψ) per Species per Habitat — Group 2 of 5
Points = posterior mean | Error bars = 95% credible interval
Greater Mouse-deer
Horse-tailed Squirrel
Large Treeshrew
Leopard Cat
Lesser Mouse-deer
Lesser Treeshrew
Long-tailed Macaque
Long-tailed Porcupine
Low's Squirrel
Malay Civet
Occupancy probability (ψ)
1.00
0.75
0.50
0.25
0.00
Trail
Clearing
Riparian
Forest
Disturbed Track
Habitat type


Occupancy Probability (ψ) per Species per Habitat — Group 3 of 5
Points = posterior mean | Error bars = 95% credible interval
Malay Porcupine
Marbled Cat
Masked Palm Civet
Moonrat
Mousedeer sp.
Muntjac sp.
Orangutan
Pig-tailed Macaque
Plaintain Squirrel
Prevost's Squirrel
Occupancy probability (ψ)
1.00
0.75
0.50
0.25
0.00
Trail
Clearing
Riparian
Forest
Disturbed Track
Habitat type


Occupancy Probability (ψ) per Species per Habitat — Group 4 of 5
Points = posterior mean | Error bars = 95% credible interval
Rat sp.
Red Muntjac
Sabah Giant Rat
Sambar Deer
Short-tailed Mongoose
Squirrel sp.
Striped Treeshrew
Sun Bear
Sunda Clouded Leopard
Sunda Pangolin
Occupancy probability (ψ)
1.00
0.75
0.50
0.25
0.00
Trail
Clearing
Riparian
Forest
Disturbed Track
Habitat type

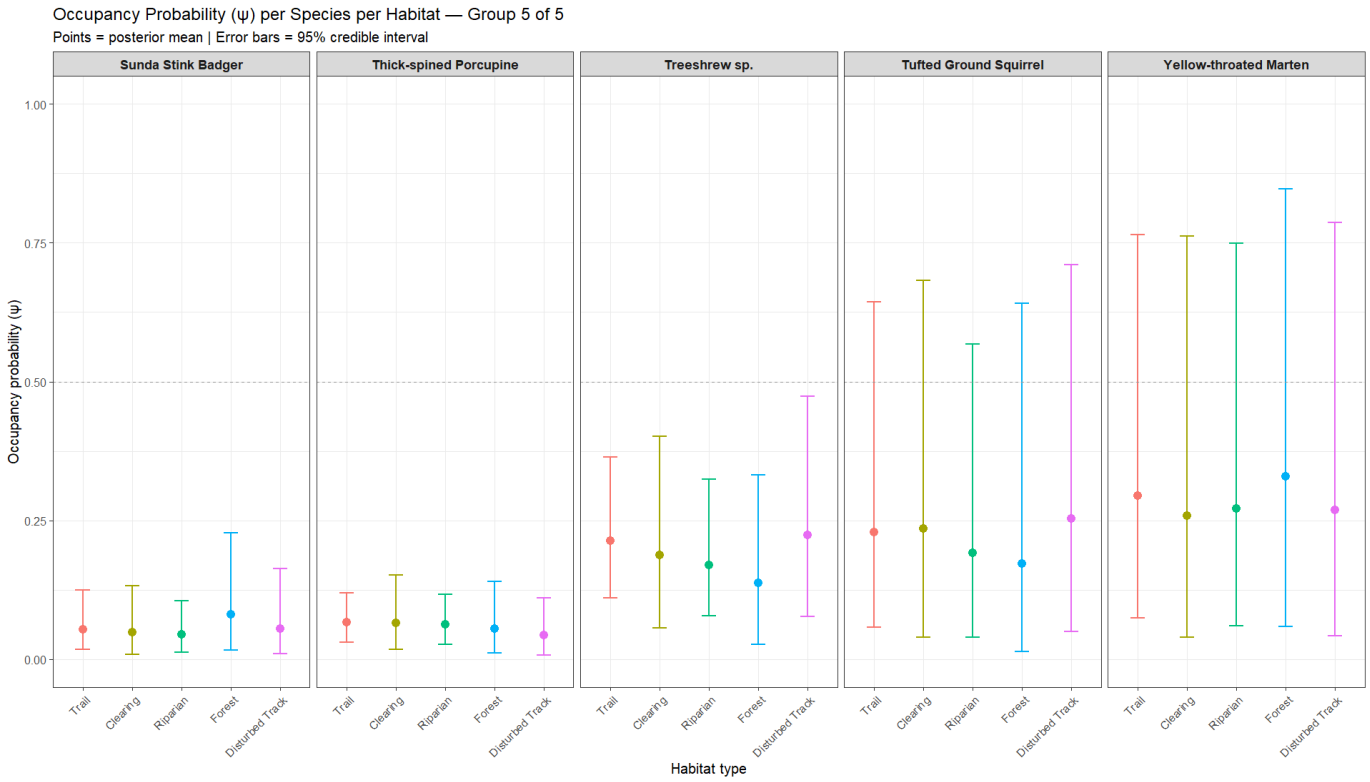


Figure 1: Occupancy probability (ψ) per species per habitat type. Points show the posterior mean and error bars show the 95% credible interval. The y-axis is standardised to $[0, 1]$ across all panels.

5.2 Detection Probability per Species

Figure 2 shows the posterior mean detection probability $\hat{p}_k$ and 95% credible interval for each species, ranked from highest to lowest detection. Detection is modelled as a species-specific constant, so each species receives a single estimate regardless of habitat.

Detection probabilities vary considerably across species, ranging from approximately 0.50 for Bearded Pig at the top of the ranking down to below 0.05 for Low's Squirrel and Sunda Clouded Leopard at the bottom. This variation is biologically meaningful, large-bodied, terrestrial species such as Bearded Pig, Greater Mouse-deer, Thick-spined Porcupine, and Lesser Mouse-deer have the highest detection probabilities, likely because they move at ground level and reliably trigger camera sensors. Small arboreal squirrels, cryptic carnivores such as Marbled Cat and Sunda Clouded Leopard, and rare species such as Banded Linsang show the lowest detection probabilities.

Notably, no species exceeds the 0.50 dashed reference line on its posterior mean, indicating that for every species in the dataset, a camera trap has less than a 50% chance of recording the species on any given week even when it is present. This underlines the importance of the occupancy modelling

framework, raw detection counts across the 45 species would substantially underestimate true occupancy without correction for this imperfect detection.

Credible intervals are generally wider for rare species, again reflecting the limited data available to estimate detection for species seen infrequently. The wide interval for Collared Mongoose, Lesser Treeshrew, and Banded Linsang in particular spans much of the 0 to 0.5 range, indicating substantial uncertainty in their detection probability estimates.

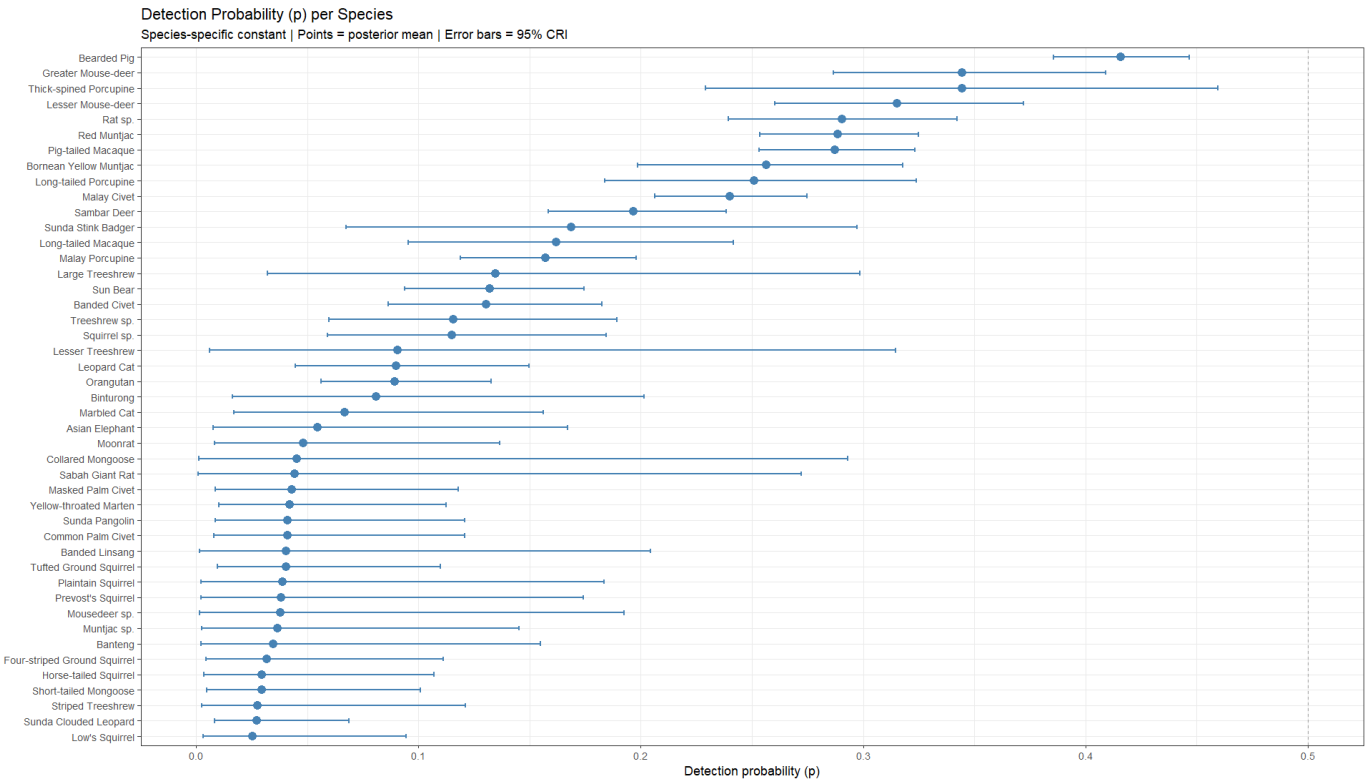


Figure 2: Detection probability (p) per species, ranked from highest to lowest.

5.3 Community-Level Mean Occupancy by Habitat

Figure 3 shows the community-level mean occupancy probability averaged across all 45 species for each habitat type. The posterior means are very similar across all five habitats, all sitting close to 0.25. Trail has the highest community mean occupancy, followed by Clearing, Forest, Disturbed Track, and Riparian in close succession.

However, all five credible intervals are very wide and substantially overlapping, extending from near zero to approximately 0.5 to 0.65. This means the model does not provide strong evidence that the community as a whole prefers any one habitat over another. The uncertainty arises from

the combination of two factors: species vary considerably in their individual habitat responses, and many species have wide individual credible intervals. When these are averaged together the community mean retains substantial uncertainty.

The Riparian habitat shows the narrowest credible interval of the five, with the upper bound reaching only to approximately 0.5. This may reflect the greater number of Riparian deployments in the dataset providing more stable estimates, or it may indicate that the mammal community is more consistently moderate in its use of riparian habitats with less extreme variation across species.

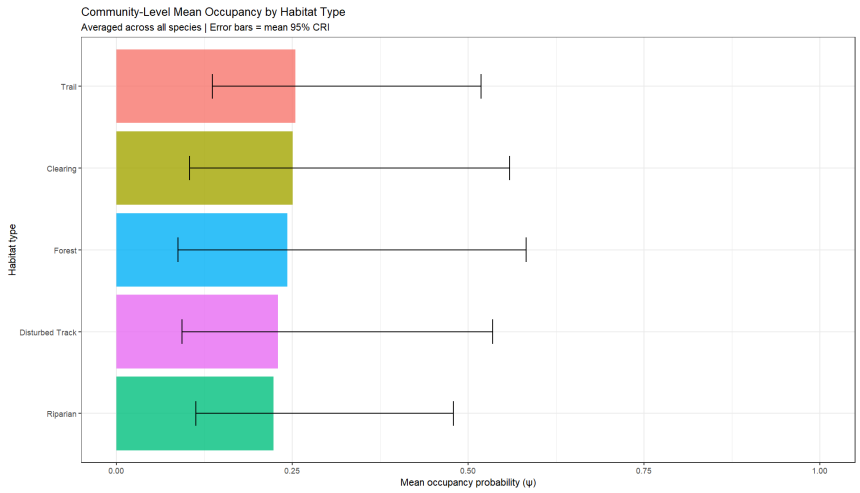


Figure 3: Community-level mean occupancy probability by habitat type.

5.4 Occupancy Probability Heatmap

Figure 4 provides the most complete overview of habitat use across the full community. Each cell shows the posterior mean occupancy probability for one species in one habitat, with darker blue indicating higher occupancy.

The heatmap confirms that most species show light to mid-blue colouring consistently across all five habitats, consistent with the low to moderate occupancy probabilities seen in Figure 1. Several species stand out clearly with dark blue rows. Bearded Pig shows the darkest and most uniform colouring across all habitats, confirming its position as the most widely occupying species in the dataset. Malay Civet, Pig-tailed Macaque, Red Muntjac, and Malay Porcupine also show consistently darker rows across habitats. Orangutan shows a particularly interesting pattern — dark cells in Trail, Clearing, Forest, and Disturbed Track but a noticeably lighter cell in Riparian, suggesting lower estimated occupancy in water-adjacent habitats compared to other habitat types.

Leopard Cat shows a striking pattern with a distinctly darker cell in the Forest column compared to the other habitats, suggesting a preference for closed forest over more open or disturbed habitats. Collared Mongoose and Long-tailed Macaque similarly show darker Forest cells, while most squirrel and treeshrew species show uniformly light colouring indicating low occupancy across

all habitats.

At the bottom of the heatmap, Banteng, Banded Linsang, and Banded Civet show very light colouring across all habitats, consistent with the low detection probabilities seen in Figure 2 and suggesting either genuine rarity across the landscape or high uncertainty in occupancy estimates due to few detections.

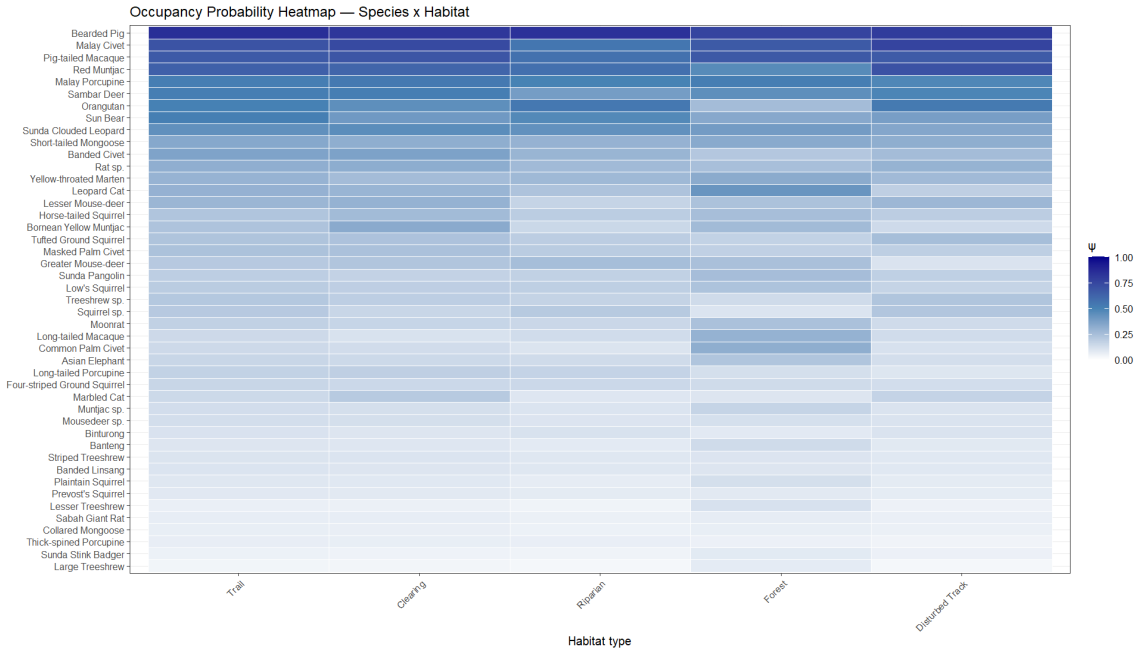


Figure 4: Occupancy probability heatmap across all species and habitat types.

6 Discussion

This section interprets the results, discusses the value of separating occupancy from detection, reflects on the modelling decisions made, and identifies limitations and directions for improvement.

6.1 Community-Level Occupancy Patterns

At the community level, the most striking finding from Figure 3 is that all five habitat types produce very similar mean occupancy estimates of approximately 0.25, with heavily overlapping credible intervals. This means the model does not provide strong evidence that the mammal community as a whole prefers any one habitat type over another.

Some species show clear habitat associations or avoidances that are visible in Figures 1 and 4, but when averaged together these opposing patterns cancel out, producing a flat community mean. Additionally, some habitat types such as Disturbed Track and Clearing had considerably fewer camera deployments than Trail and Riparian, meaning the community mean for those habitats is estimated with greater uncertainty. The wide credible intervals in Figure 3 therefore reflect both genuine ecological uncertainty and a data limitation.

6.2 Species-Level Habitat Associations

Bearded Pig stands out clearly as the most widely occupying species in the dataset, with consistently high posterior mean occupancy across all five habitats and among the narrowest credible intervals of any species. This is ecologically consistent with the known ecology of Bearded Pig in Borneo, which is documented as an animal that travels long distances across different habitat types in search of mast fruiting events.

Orangutan shows relatively high occupancy across Trail, Clearing, Riparian, and Disturbed Track, but a noticeably lower estimate in Forest habitat. This is somewhat unexpected given that Orangutans in Sabah are documented in a wide range of habitat conditions including logged forest. The lower Riparian estimate may reflect genuine avoidance of open water margins, or it may be a product of fewer Orangutan detections at forest sites, leading the model to produce a lower estimate there. This would be worth investigating with a larger dataset.

Leopard Cat shows a clear Forest preference visible in the heatmap, with a distinctly darker cell in the Forest column compared to other habitats. This is ecologically accurate because Leopard Cat is a forest-dependent carnivore whose prey base of small rodents, birds, and lizards is most abundant in forest habitat.

Malay Civet and **Pig-tailed Macaque** both show consistently high occupancy across all habitats, suggesting they are resilient generalists able to exploit degraded and disturbed habitats alongside intact forest. Both species are known to be adaptable to human-modified landscapes, and this result is consistent with that characterisation.

Species such as **Banded Linsang**, **Sunda Clouded Leopard**, and **Sunda Pangolin** show very low occupancy estimates with wide credible intervals across all habitats. For these species it is difficult to distinguish between genuine landscape-scale rarity, restricted but undetected habitat

preferences, and simply insufficient detections to produce informative posterior estimates. These species would benefit from targeted survey effort.

6.3 Detection Probability and the Value of Occupancy Modelling

Figure 2 shows that detection probability varies across species, ranging from around 0.50 for Bearded Pig down to below 0.05 for Low's Squirrel and Sunda Clouded Leopard. No species exceeds 0.50 detection probability on its posterior mean, meaning for every species in this dataset, a camera trap is more likely to miss the species than record it on any given week even when the species is present.

This finding justifies the occupancy modelling framework. If raw detection counts were used instead, species with low detection probability would appear less even if they occupy many sites. A species with $p = 0.05$ and $\psi = 0.8$, would be detected at a given site on only a small fraction of weeks even though it is nearly always there. Using detection counts for occupancy would lead to underestimating how widespread this species truly is. The occupancy model corrects for this by estimating the true probability of site use independent of how catchable each species is.

The variation in detection probability also has a clear biological explanation. Large, ground species such as Bearded Pig, Greater Mouse-deer, and Lesser Mouse-deer move predictably at ground level and reliably trigger camera sensors. Small arboreal species and cryptic carnivores such as Sunda Clouded Leopard and Marbled Cat have large home ranges and low encounter rates per site per week. These differences in detectability are precisely what the model is designed to account for.

6.4 Limitations

Habitat category merging. Consolidating 18 raw habitat labels into 5 groups was necessary to produce reliable coefficient estimates but loses potentially important distinctions. Riparian logged forest and Riparian may affect species very differently but are treated identically in the model. Similarly, Oil palm Riparian Reserve was grouped with Forest based on structural similarity to forest fragments, but this may not hold for all species.

Sample size imbalance means estimates are not equally reliable across habitat types. Trail and Riparian had substantially more camera deployments than Disturbed Track and Clearing. This imbalance means that effects in the less sampled habitats carry greater uncertainty, and real ecological differences between habitats may not be detectable with the available data.

Temporal pooling treats all detections from 2014 to 2017 as coming from a single static occupancy state per species per site. In reality, species may have colonised or vacated sites across the four years as the logged landscape continued to change following ongoing timber extraction and natural forest recovery. Pooling across years assumes the system is in a stable state, which is unlikely to be true in an actively managed logging concession.

6.5 Improvements

A **dynamic multi-season occupancy model** using the four years of data separately would estimate colonisation and local extinction probabilities for each species across habitat types. This

would directly quantify whether the mammal community is responding to ongoing changes in the SAFE landscape.

Including **additional covariates** such as elevation, distance to forest edge, or canopy cover could explain residual variation in occupancy not captured by the broad habitat categories used here. These continuous variables would allow more habitat associations to be estimated and would help identify whether the habitat effects observed are driven by the habitat label itself or by correlated environmental gradients such as altitude or disturbance intensity.

Finally, a **species interaction model** would test whether the occupancy of predator species such as Leopard Cat is associated with changes in the occupancy of their prey species at the same sites. The current model treats each species independently within the community hierarchy, but ecological interactions between species may be important for occupancy patterns that this model cannot detect.

7 Conclusion

This project set out to investigate whether habitat type affects the occupancy and detection of mammal species in a logged forest in Borneo, using camera trap data collected across 260 sites between 2014 and 2017. A Bayesian multi-species occupancy model was fitted to data from 45 mammal species across five habitat types which are trail, clearing, riparian, forest, and disturbed track, using the `msPGOcc` function from the `spOccupancy` package in R. The model simultaneously estimated occupancy probability as a function of habitat type and a species-specific detection probability, with all species modelled jointly through a hierarchical community-level prior structure.

At the community level, mean occupancy probability was approximately 0.25 across all five habitat types with substantially overlapping credible intervals, indicating no single habitat dominates mammal occupancy across the community as a whole. At the species level, Bearded Pig was the most widespread species across all habitats, consistent with its known landscape-scale movement behaviour in Borneo. Leopard Cat showed a clear preference for Forest habitat, supporting the conservation value of maintaining forest fragments in logged landscapes. No species exceeded a detection probability of 0.50, confirming that camera trap data without occupancy modelling would substantially underestimate the true distribution of every species in the dataset.

Several limitations remain, including the merging of habitat categories which loses ecological nuance or the pooling of data across four years without accounting for temporal change. Future extensions including a dynamic multi-season model, and the inclusion of additional environmental covariates such as elevation would address these limitations and provide a more complete picture of how mammal communities respond to habitat disturbance in logged tropical forests. Ultimately, this project demonstrates that rigorous statistical modelling of camera trap data can provide relevant information about species-habitat relationships in threatened landscapes, even when detection is imperfect and many species are rarely observed.

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